



Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Term 1, 2019

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED: 2 hours

TOTAL MARKS: 100

TOTAL NUMBER OF QUESTIONS: 5

THIS EXAM CONTRIBUTES 50% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 6 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

- a. (10 marks) For the circuit shown in Figure 1,
- (7 marks) Use nodal analysis to find voltage v_0 .
 - (3 marks) Find the power supplied/absorbed by the dependent voltage source.

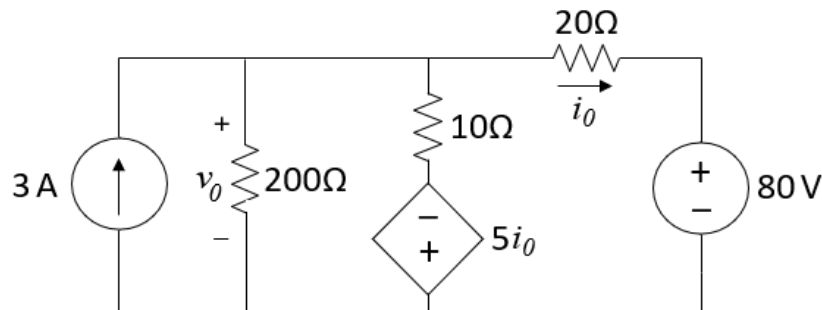


Figure 1

- b. (10 marks) The circuit shown in Figure 2 is being used to power a hot water heater. The heater can be modelled as a single resistor.
- (4 marks) Calculate the Thevenin voltage at terminals $a-b$.
 - (4 marks) Calculate the heater resistance that will ensure the maximum transfer of power from the circuit to the heater.
 - (2 marks) Find the maximum power that can be delivered to the heater.

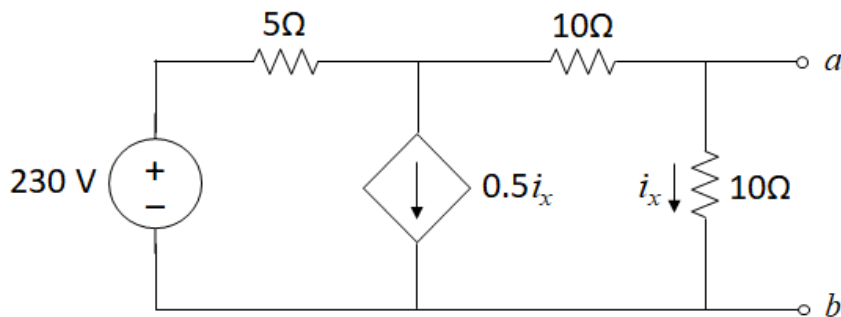


Figure 2

QUESTION 2 [20 marks]

- a. (8 marks) Find voltage v , current i , and the energy stored in the capacitor in the circuit of Figure 3 under DC conditions.

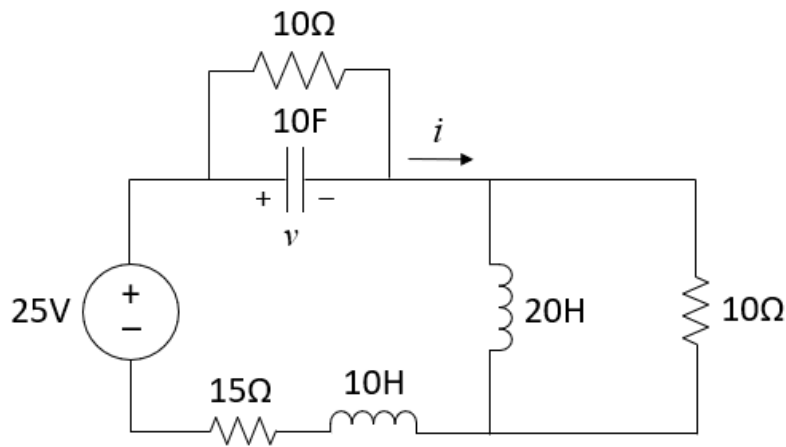


Figure 3

- b. (12 marks) The circuit below is used to take signals from a Hi-Fi system and amplify them so that they can drive a subwoofer. When first turned on, the amplifier fails and applies +15V DC directly to the output (v_a). After 30 milliseconds, the speaker protection circuit operates Sw1 to disconnect the speaker and connect it to ground (i.e., Sw1 changes from position '1' to position '2').

Considering that the amplifier was turned on at $t=0$ seconds, calculate the voltage across the capacitor, $v_c(t)$, for the first 60 milliseconds (i.e., for $0 < t \leq 30$ ms and $30 \text{ ms} < t \leq 60$ ms).

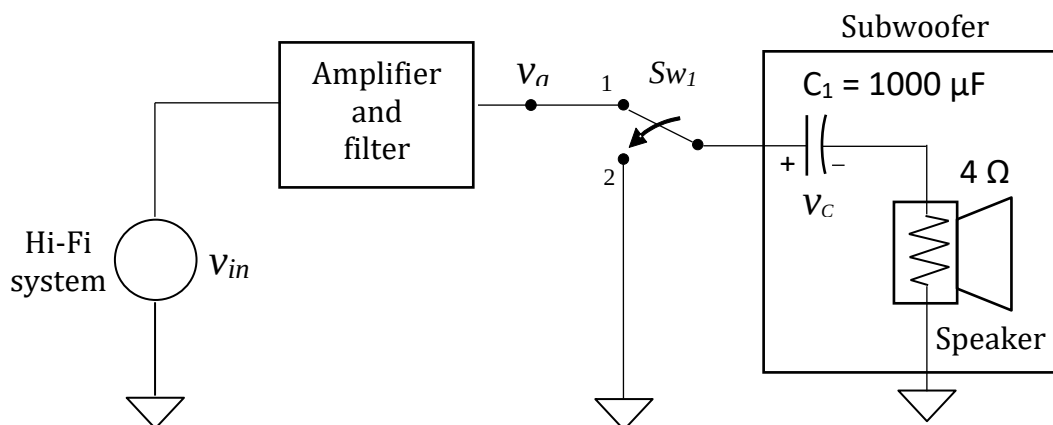


Figure 4

QUESTION 3 [25 marks]

- a. (12 marks) The two channels of a stereo amplifier are fed the same mono audio signal, v_s . A speaker is connected between the two amplifier outputs, bridging the output terminals, as shown in Figure 5. This circuit, known as bridge amplifier, is used to increase the amount of power available in the speaker. Assuming ideal op amps:
- (4 marks) Derive an expression for the top op-amp output voltage v_1 in terms of the input voltage v_s .
 - (4 marks) Derive an expression for the bottom op-amp output voltage v_2 in terms of the input voltage v_s .
 - (4 marks) Derive an expression for the voltage gain v_0/v_s .

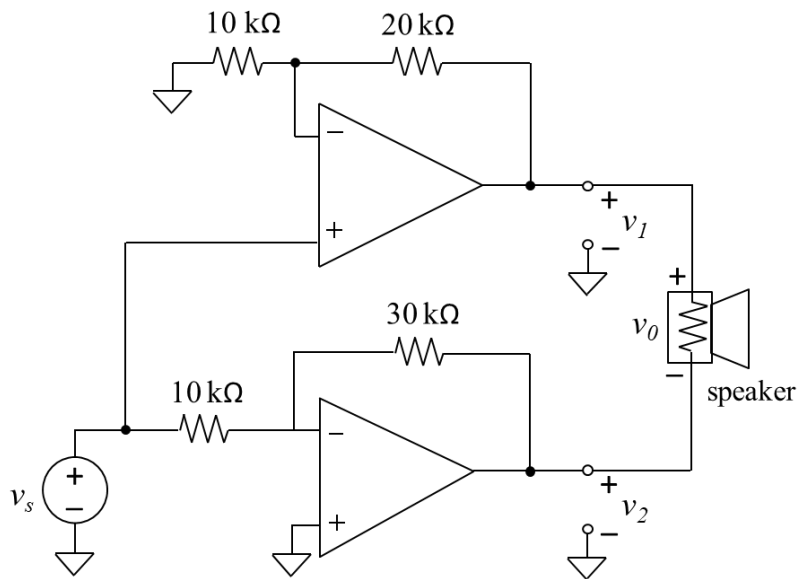


Figure 5

- b. (13 marks) Calculate V_0 in the circuit shown in Figure 6.

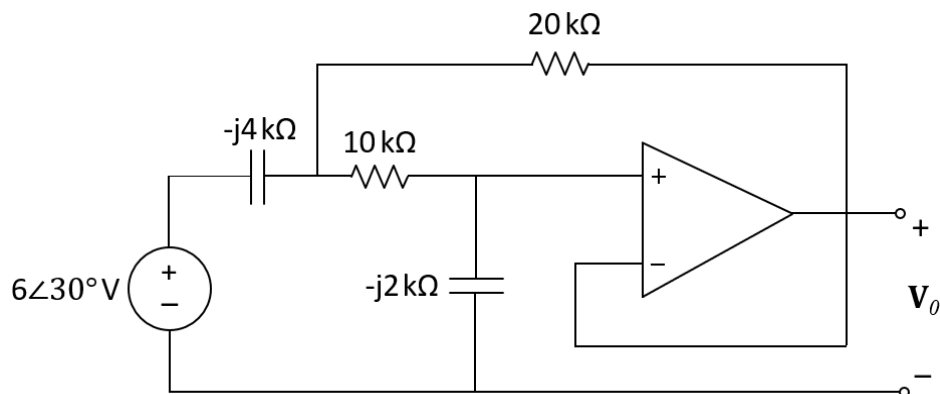


Figure 6

QUESTION 4 [25 marks]

- a. **(13 marks)** The circuit shown in Figure 7 corresponds to a “woofer” speaker which reproduces only the lower frequencies of music. The circuit diagram for the system is shown below.
- (3 marks)** Express the voltage at the amplifier output v_a as a function of time and as a phasor.
 - (6 marks)** Find the voltage v_{out} across the speaker as a phasor.
 - (4 marks)** Find the average power consumed by the $1\text{-}\Omega$ internal resistance of the filter inductor.

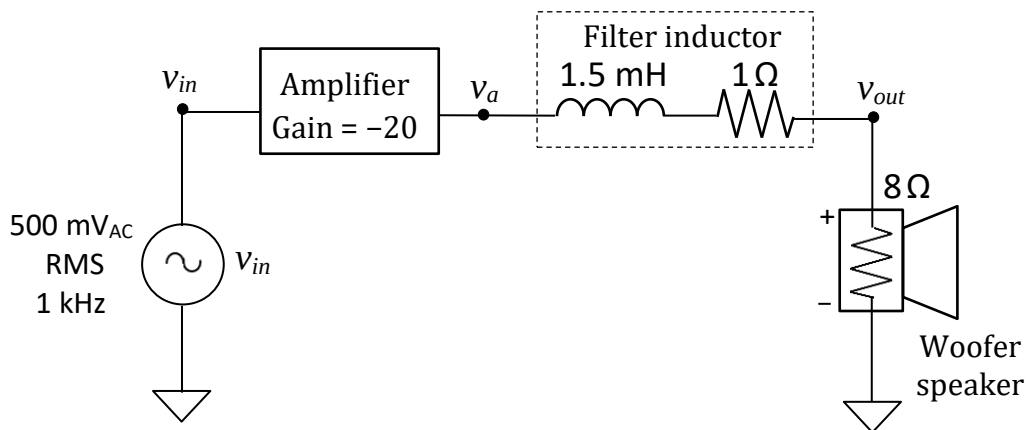


Figure 7

- b. **(12 marks)** In the circuit of Figure 8, determine the average power absorbed by the $40\text{-}\Omega$ resistor.

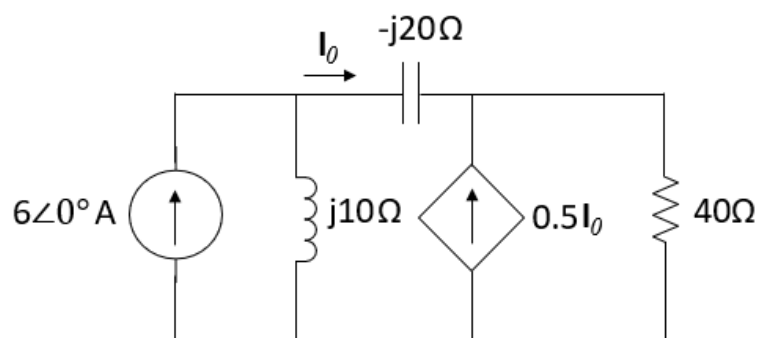


Figure 8

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QUESTION 5 [10 marks]

- a. (2 marks) Given the logic diagram in Figure 9, derive the logical expression for Z.

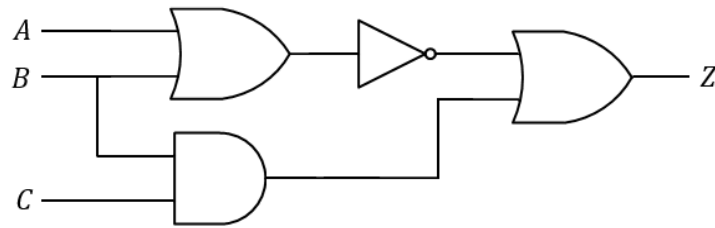


Figure 9

- b. (8 marks) Consider the following identity:

$$\overline{(A + B) \cdot (\bar{A} + A \cdot B)} = \bar{B}$$

- (5 marks) Prove the identity using a truth table.
- (3 marks) Draw the logic diagram which represents the first term of the identity (i. e. $\overline{(A + B) \cdot (\bar{A} + A \cdot B)}$).

END OF PAPER